

Bhautik R. Vaghamshi

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Ahmedabad, India

Education

GLS University – Faculty of Engineering & Technology

Ahmedabad, India

B.Tech in Computer Science & Engineering | GPA: 3.0/4.0

Aug 2023 – May 2027

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Statistical Methods, Computer Networks

Publication

Modern Learning Paradigms Beyond Traditional Supervision | *ScienceOpen Preprints*

Nov 2025

- Survey on self-supervised, semi-supervised, few-shot, zero-shot, and meta-learning paradigms across CV and NLP.
- DOI: 10.14293/PR2199.002416.v1

Technical Projects

Agency Workflow Automation Platform | *Python, Flask, LangChain, Gemini API, FAISS*

Team

- Built an AI-assisted agency ops tool automating lead outreach, reply classification, meeting scheduling, and proposal drafting.
- Used Gemini API and FAISS-based RAG to personalize emails from past client context; integrated Gmail IMAP/SMTP and Google Calendar.
- Added human approval gates before all outbound actions; deployed dashboard views with SQLAlchemy models.

MiniGPT: Character-Level Language Modeling | *PyTorch, Transformers*

Solo

- Implemented a GPT-style transformer from scratch with self-attention for character-level text generation on Tiny Shakespeare.
- Built and benchmarked RNN, LSTM, and Transformer architectures; experimented with LR schedules, context lengths, and embedding sizes.

CLIP-Based Visual Question Answering | *PyTorch, CLIP, CLEVR*

Solo

- Developed a VQA system combining CLIP image/text embeddings with multimodal fusion and gating mechanisms for answer prediction.
- Compared two model architectures with different hidden dimensions and training strategies; analyzed trade-offs in performance.

Conditional Generative Models (GAN & CVAE) | *PyTorch, Deep Learning*

Solo

- Implemented GANs for image generation and CVAEs for class-conditioned MNIST synthesis; studied adversarial training dynamics.
- Extended to a text-conditioned CVAE generating geometric shapes from natural-language prompts.

Financial Management System | *Next.js, Supabase, Shadcn/ui, Tailwind CSS*

Solo

- Built a personal finance web app for transaction tracking, budgets, and recurring payments with analytics dashboards.
- Implemented Supabase authentication and row-level security; designed responsive UI with shadcn/ui components.

Leadership & Entrepreneurship

Co-Founder & Technical Lead

Remote

Colrnx – AI-Verified Hiring Platform

- Co-founded Colrnx, a “Proof of Work” hiring platform that verifies candidate skills through real project contributions.
- Awarded **Rs. 11,000 grant**; accepted into the **/function1 Founders Hub**. Platform live at colrnx.com (Beta).

Technical Skills

Languages: Python, JavaScript/TypeScript, C/C++, SQL

ML & AI: PyTorch, Hugging Face Transformers, Scikit-learn, FAISS, LangChain, CNNs, RNNs, Transformers, GANs, VAEs

Web: React.js, Next.js, Node.js, REST APIs, Tailwind CSS, Shadcn/ui

Databases & Tools: PostgreSQL, Supabase, MySQL, Git, Docker

Certifications

- **Introduction to Machine Learning** – NPTEL (May 2025): 8-week course on supervised/unsupervised learning and model evaluation
- **Recommender Systems** – NPTEL (May 2024): Collaborative filtering, content-based filtering, and hybrid algorithms
- **AI and Green Skills** – Edunet Foundation (July 2025): Sustainable AI practices and environmental considerations